BIOC0001

Lecture 1

Introduction to Cell Biology

Dr. Suzanne Ruddy

Texts used in these lectures:

Molecular Cell Biology, Lodish et al

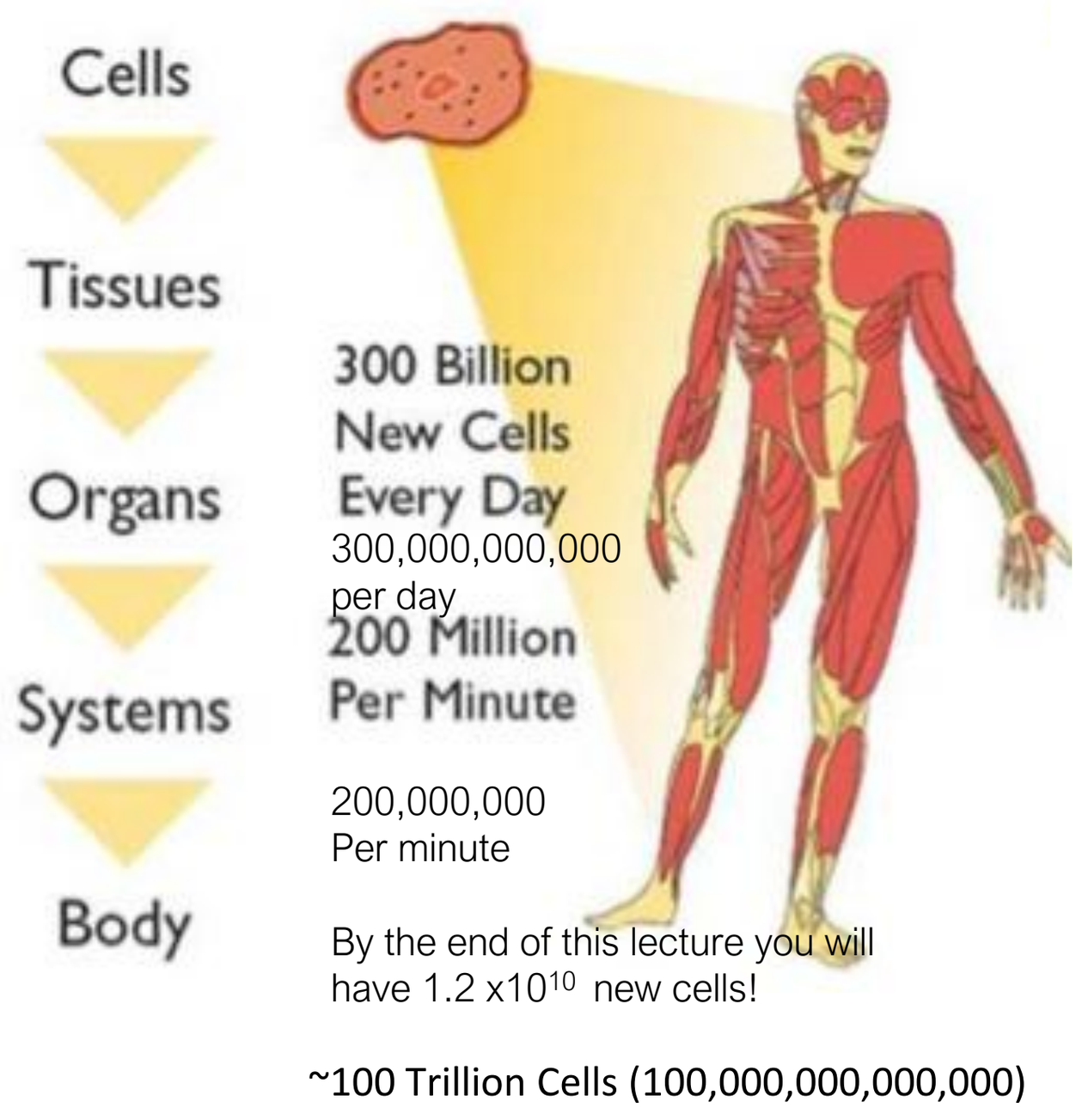
Principles of Biochemistry, Horton et al

Cells

- The cell is the basic unit of life

- Simple organisms can consist of a single cell

- Living organisms can be viewed as a collection of cells organised in specialised structures (*e.g.* organs)



~100 Trillion Cells (100,000,000,000,000)

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Cells

All organisms are related and most probably evolved from a common, single-celled ancestor.

They can be arranged in an evolutionary tree (species/organism tree).

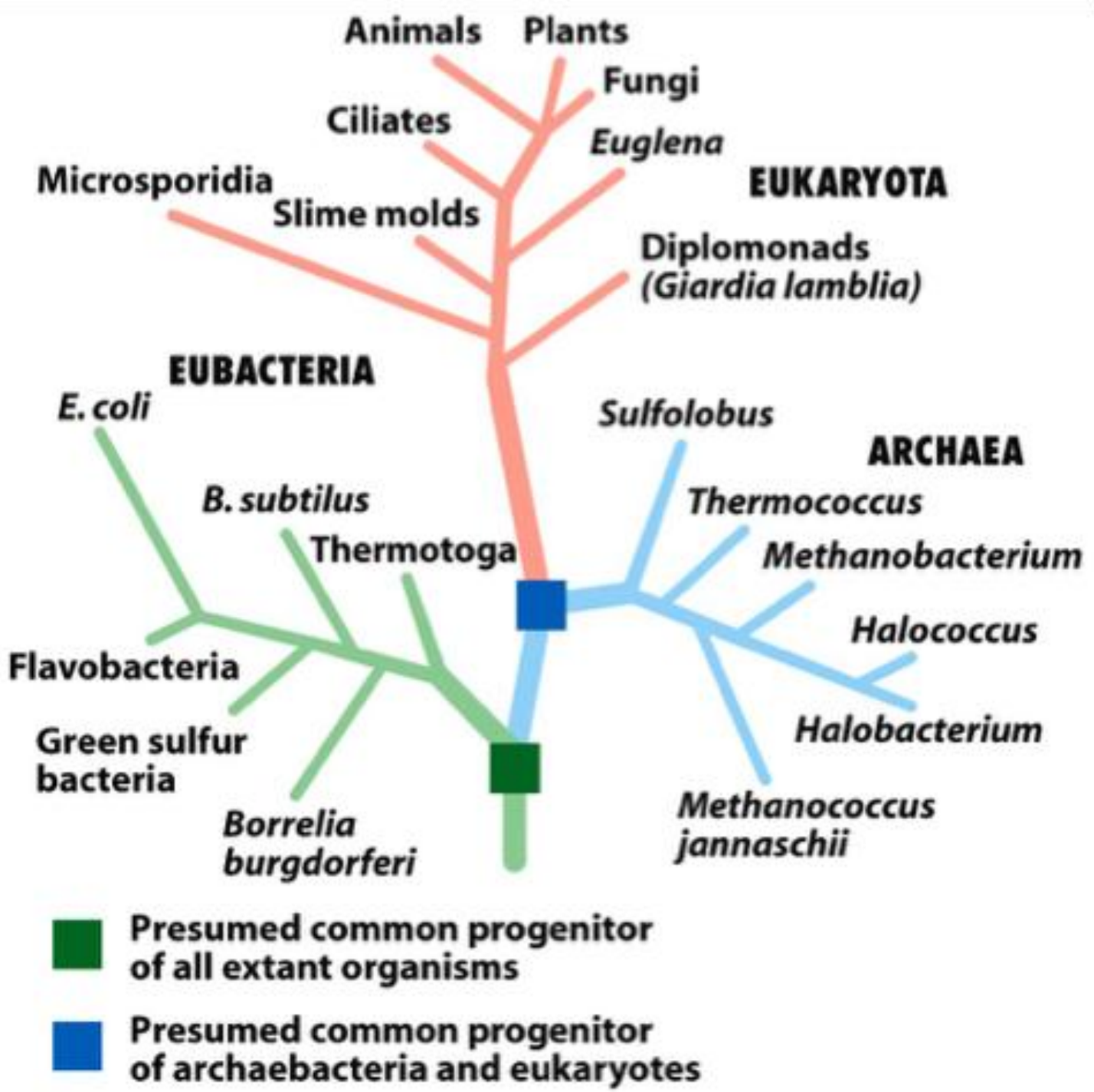


Figure 1-3
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Key cellular components are conserved throughout evolution.
The tree below shows the conservation of a molecule, the globin protein.
This is a gene tree, not a species tree.

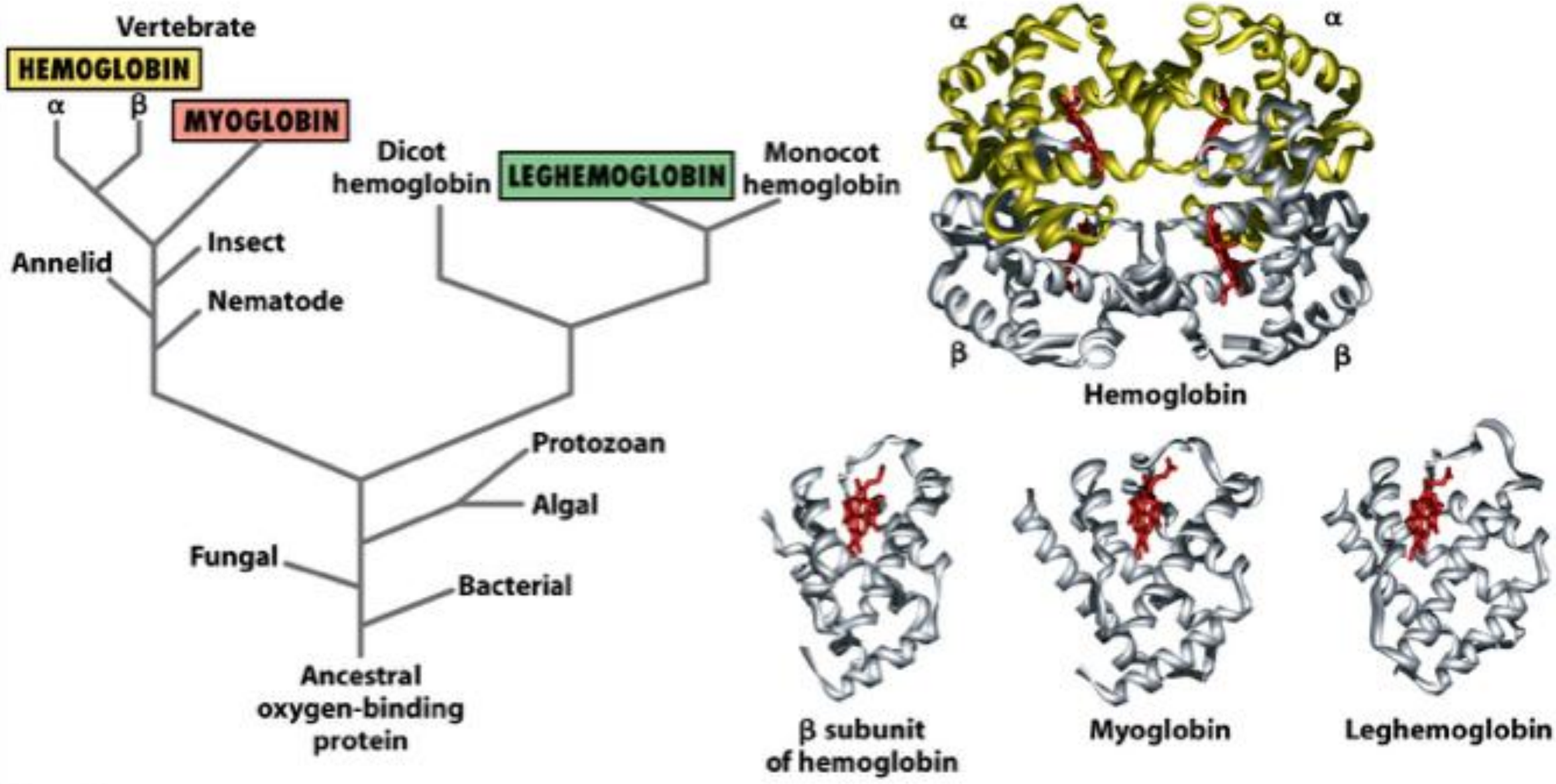


Figure 3-13
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Key motifs are conserved throughout evolution.

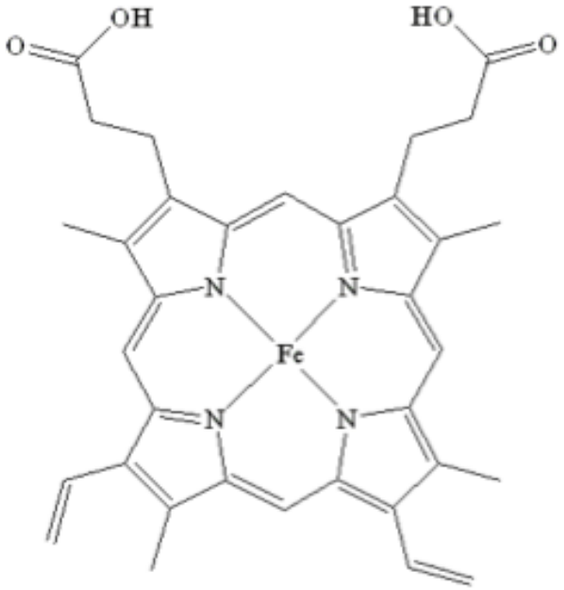
These key molecules share a similar structure (porphyrin ring).

Different metals are coordinated (Fe^{2+} vs Mg^{2+}).

Prosthetic group:

- Non-peptide (protein) group that binds covalently or non-covalently
- Can be metals or organic
- Acts to support structure of the protein
- Can also act as an electron carrier

Haem



porphyrin ring

Pyrrol ring

Methine group (CH)

Chlorophyll

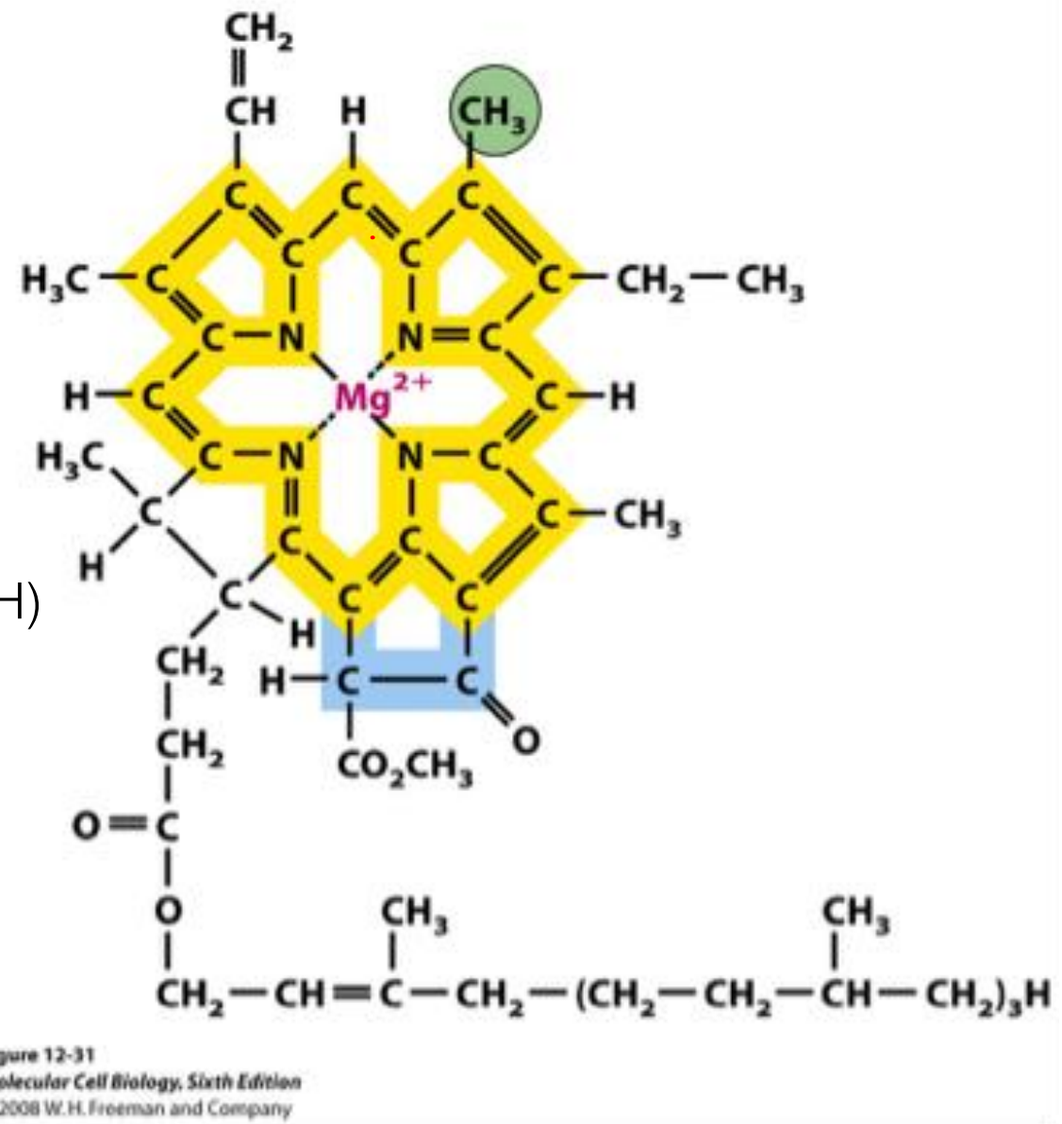


Figure 12-31
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Cells

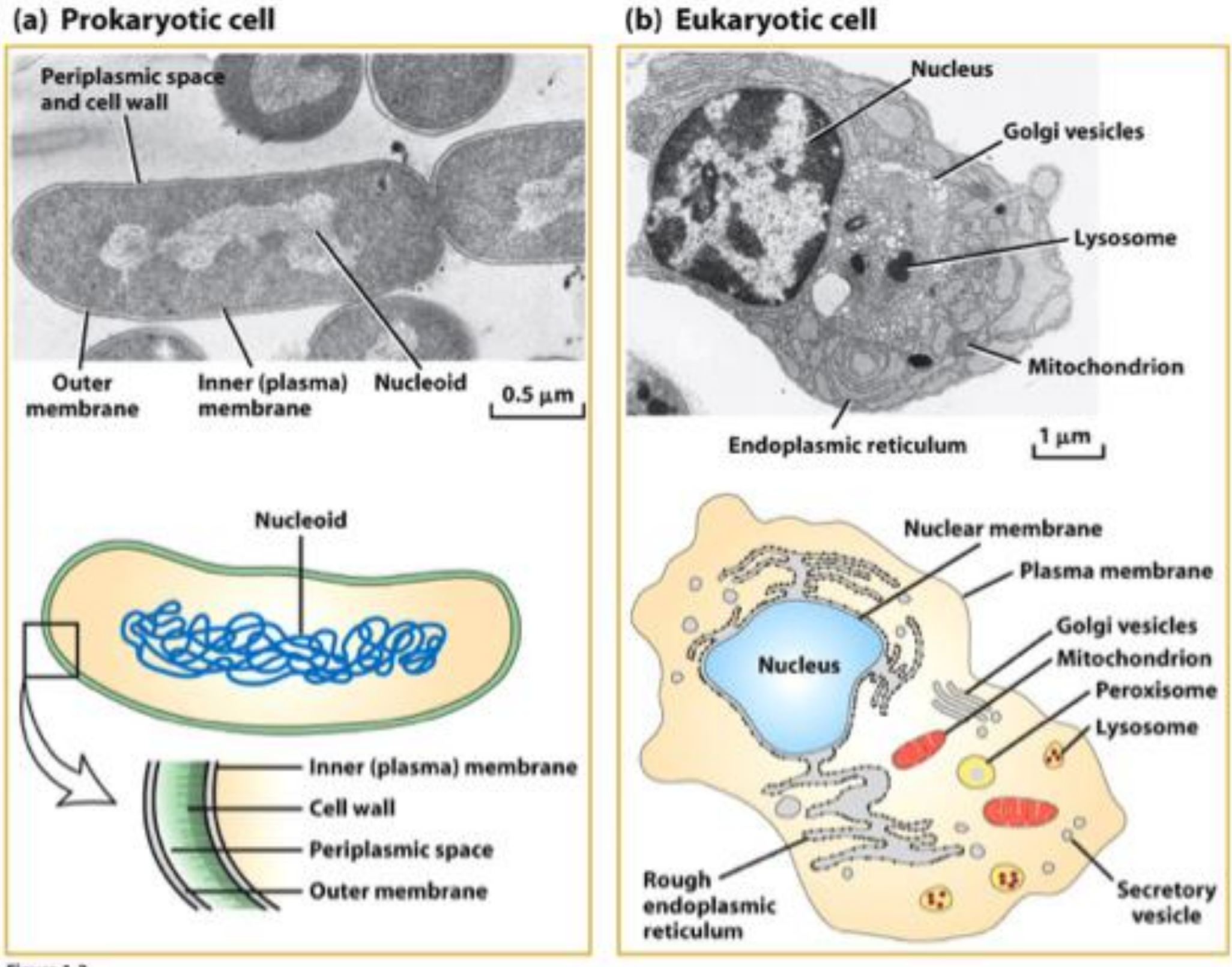


Figure 1-2
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Cellular functions: organelles

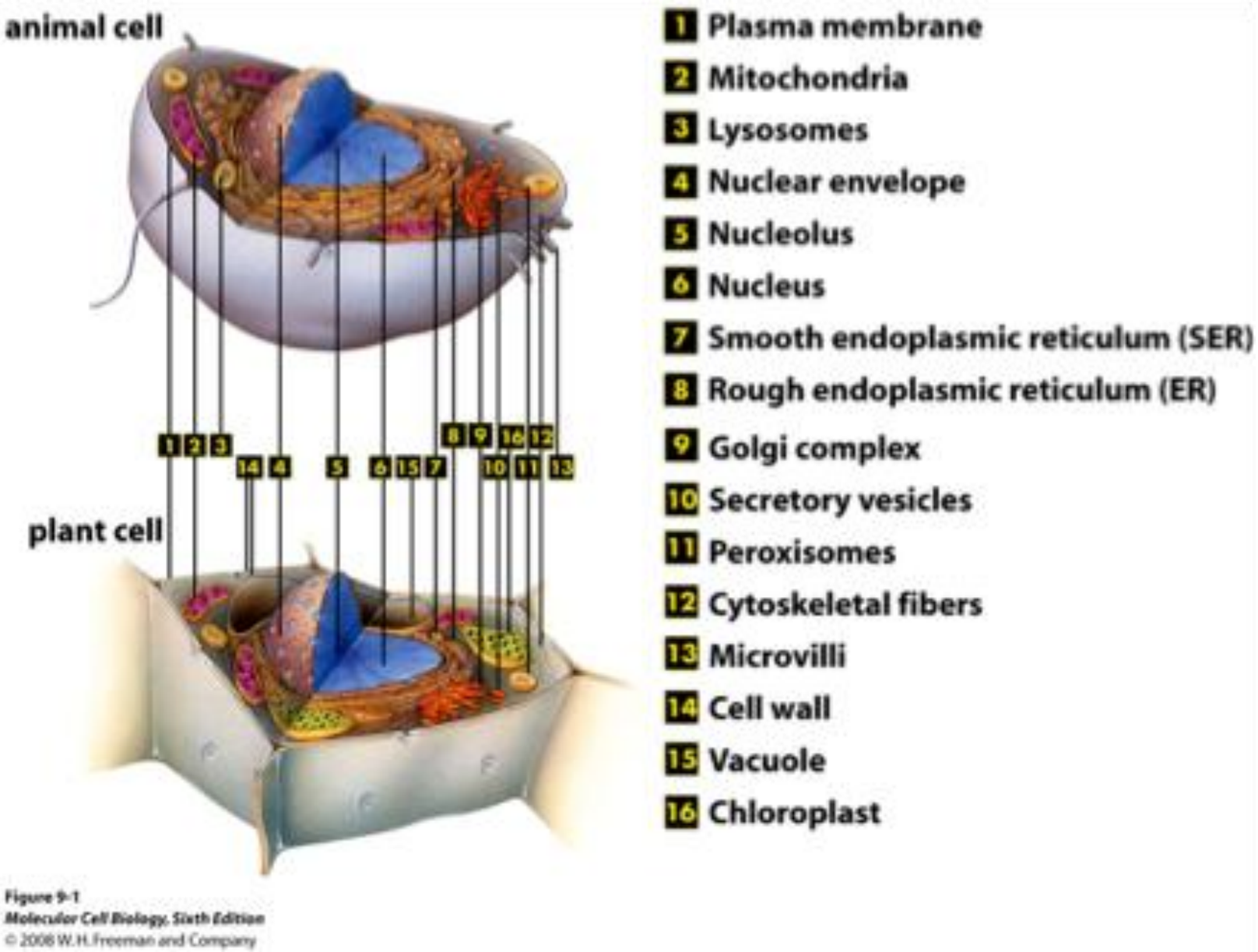


Figure 9-1
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Cellular compartments-only in eukaryotes

The plasma membrane defines the cell.
The different functions are compartmentalised within the cell in organelles.
Membranes are used to define and maintain the compartmental boundaries.

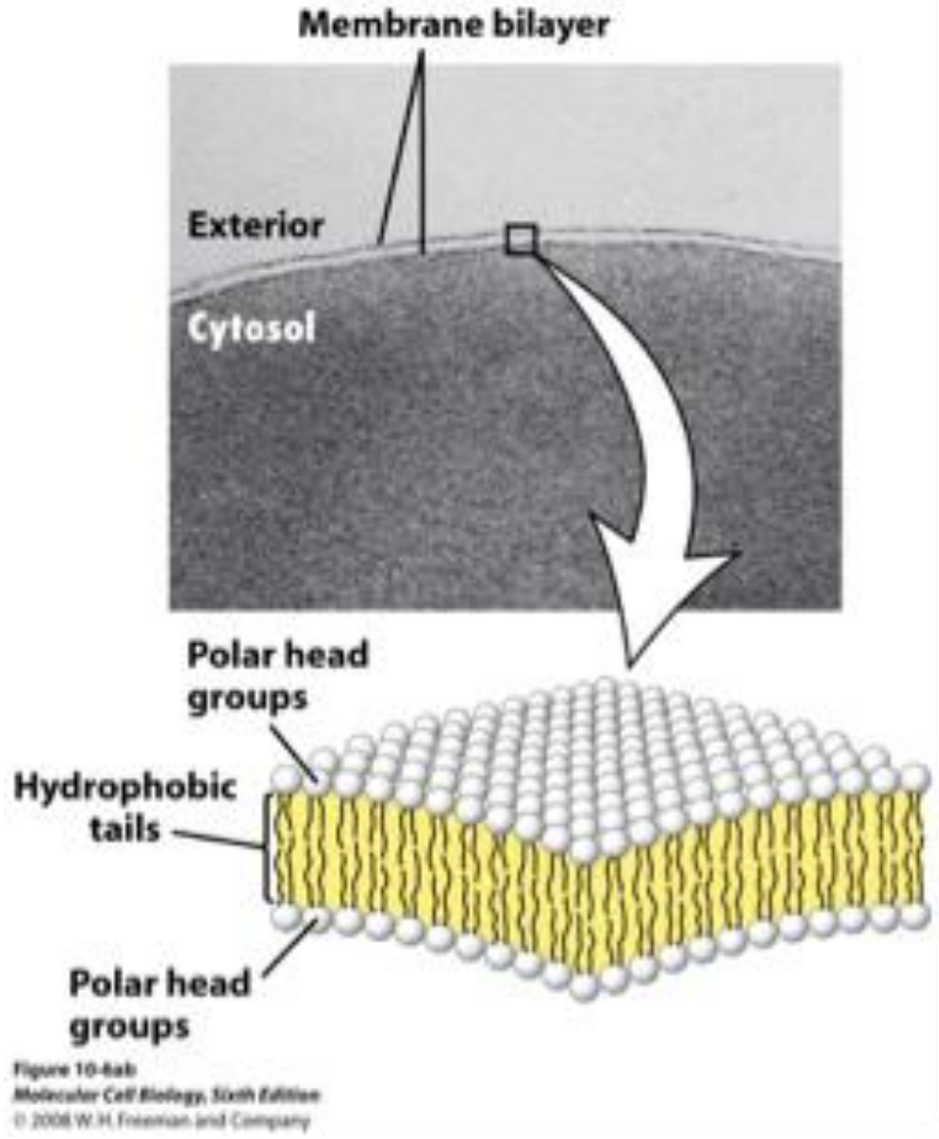


Figure 10-4ab
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Each organelle and all vesicles have their own distinctive membranes

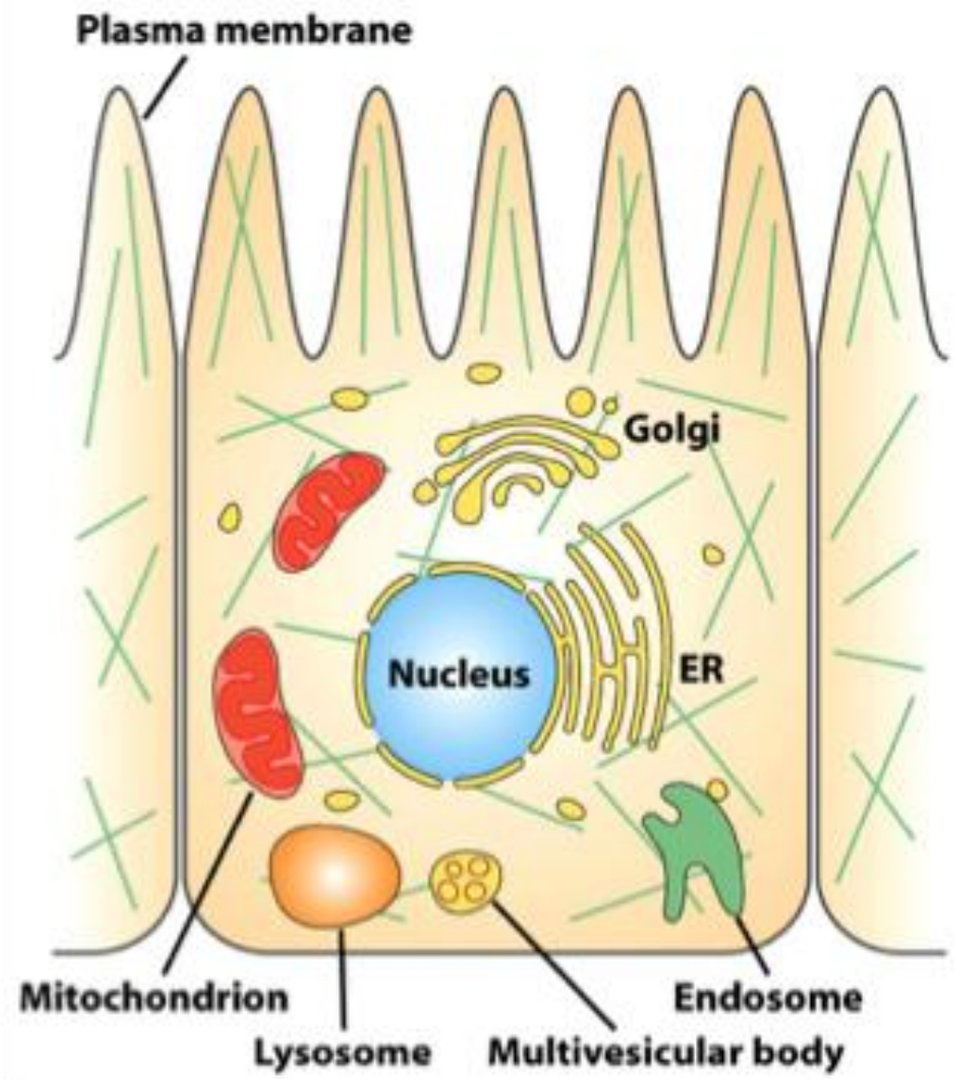


Figure 10-2
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Cellular compartments

The membrane bilayer:

- contents relate to the function of the compartment (varying lipid proportions and lipid anchored proteins).
- controls permeability, permitting selective transport of materials in and out of the compartment.

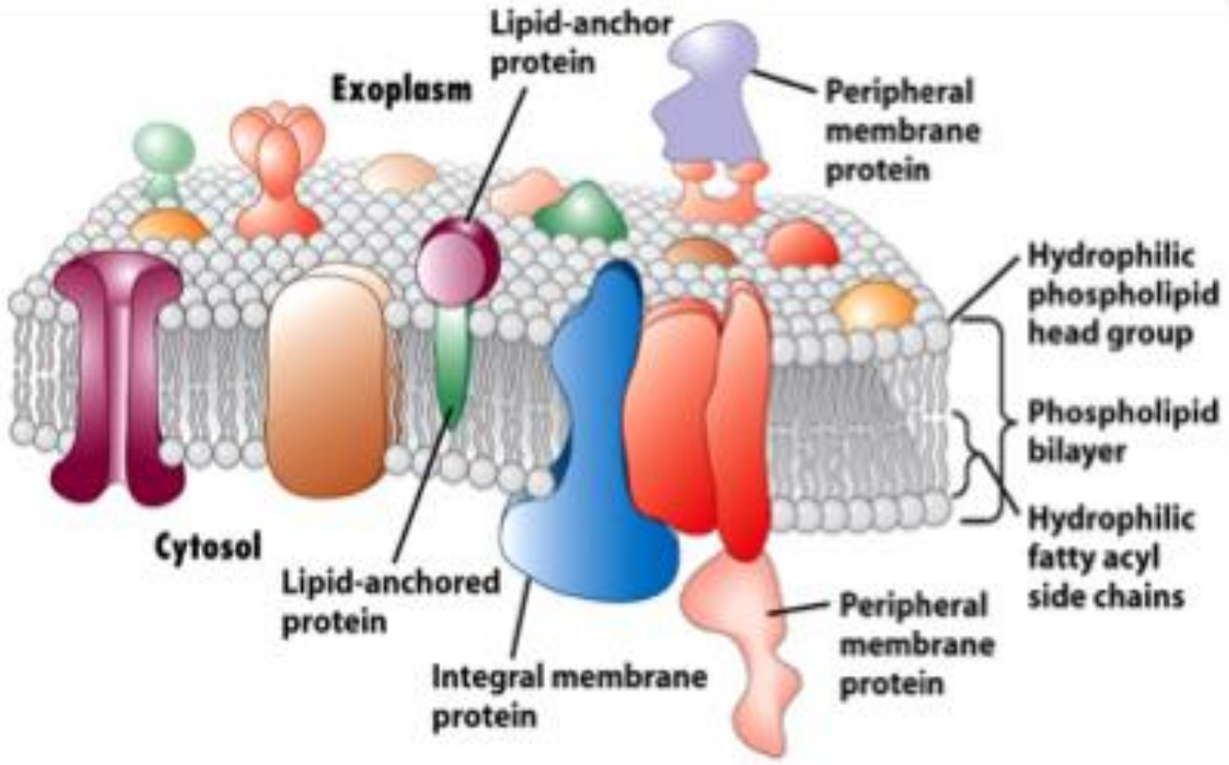


Figure 10-1
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Typo alert!

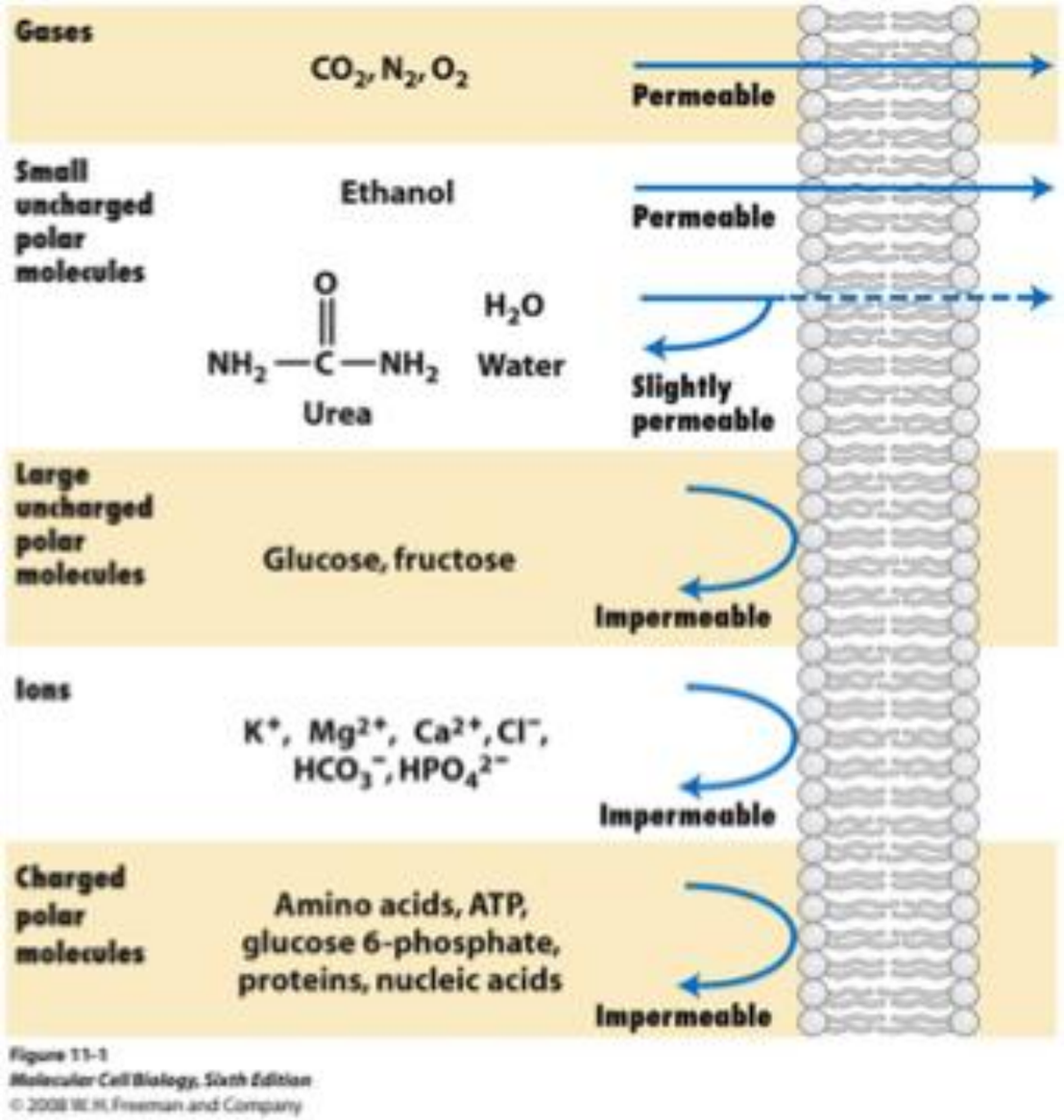


Figure 11-1
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Cellular compartments

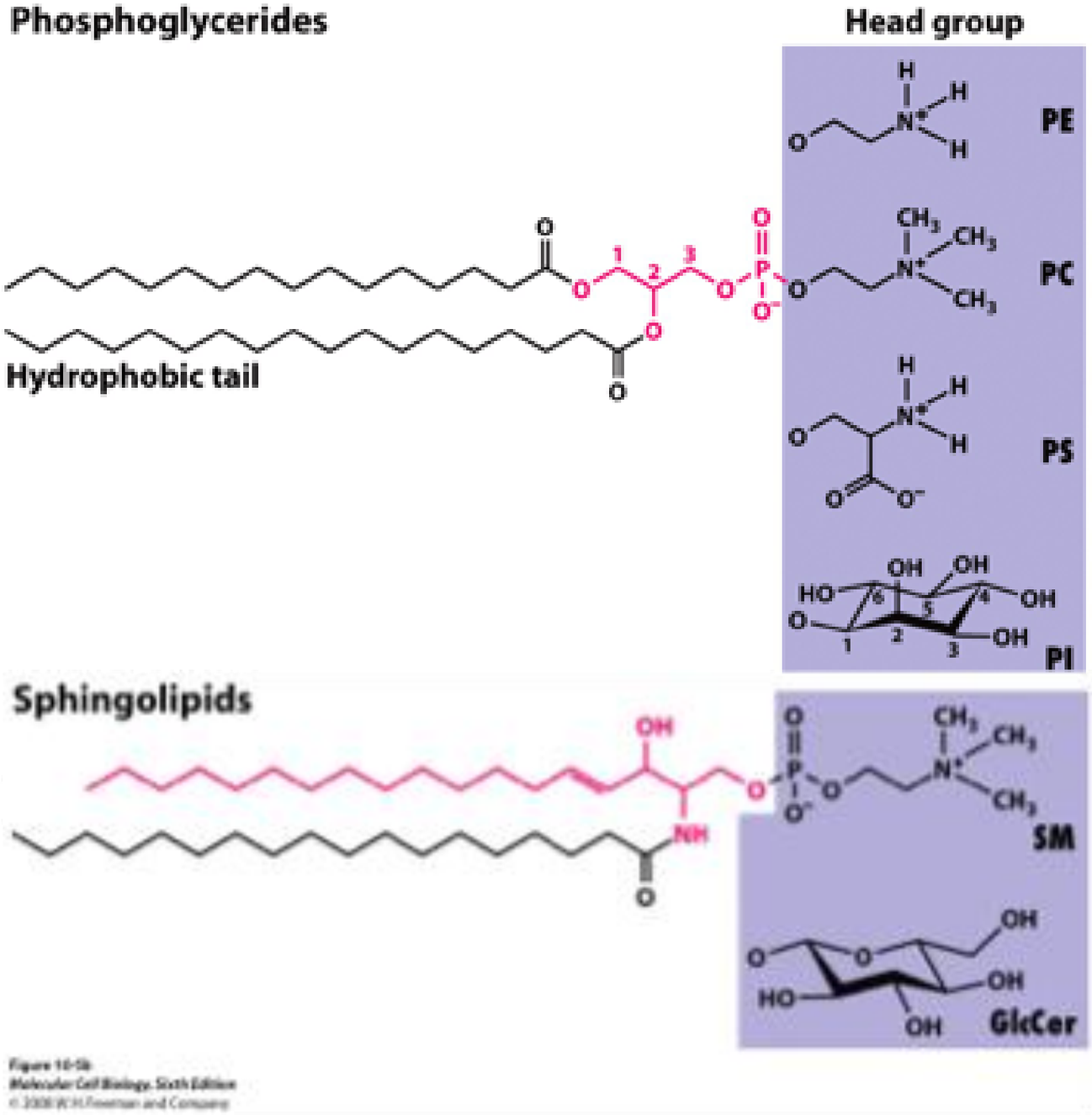
TABLE 10-1 Major Lipid Components of Selected Biomembranes

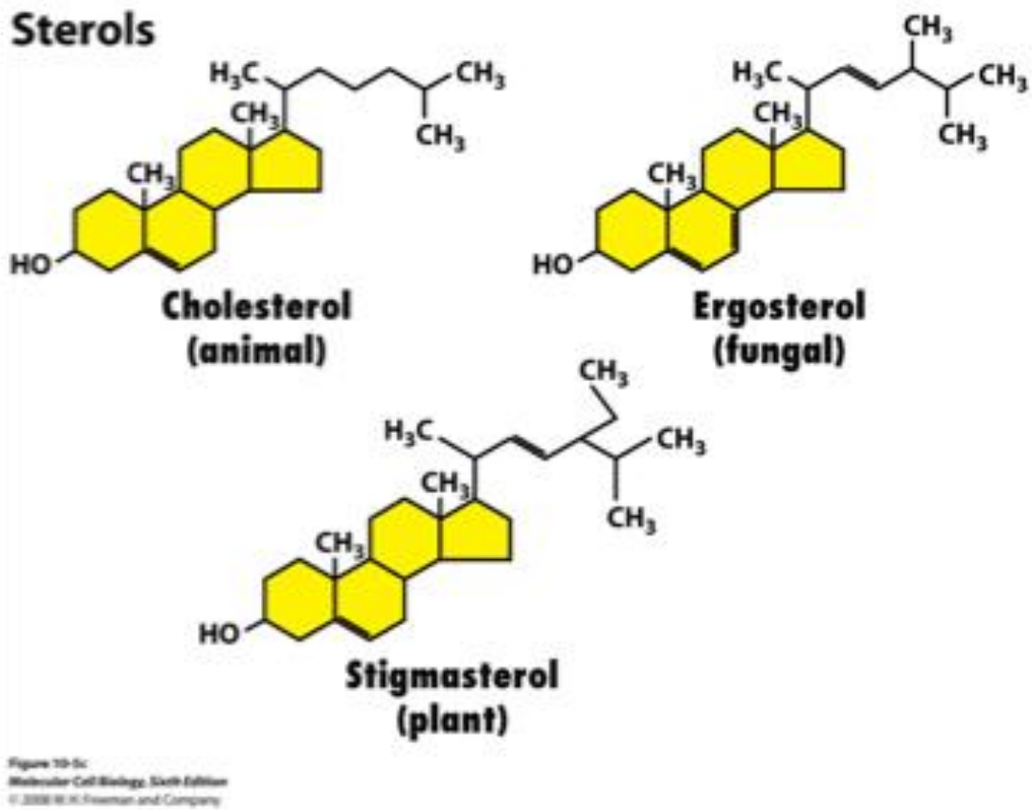
	COMPOSITION (MOL %)			
SOURCE/LOCATION	**PC**	**PE + PS**	**SM**	**CHOLESTEROL**
Plasma membrane (human erythrocytes)	21	29	21	26
Myelin membrane (human neurons)	16	37	13	34
Plasma membrane (*E. coli*)	0	85	0	0
Endoplasmic reticulum membrane (rat)	54	26	5	7
Golgi membrane (rat)	45	20	13	13
Inner mitochondrial membrane (rat)	45	45	2	7
Outer mitochondrial membrane (rat)	34	46	2	11
Primary leaflet location	Exoplasmic	Cytosolic	Exoplasmic	Both

PC = phosphatidylcholine; PE = phosphatidylethanolamine; PS = phosphatidylserine; SM = sphingomyelin.
SOURCE: W. Dowhan and M. Bogdanov, 2002, in D. E. Vance and J. E. Vance, eds., *Biochemistry of Lipids, Lipoproteins, and Membranes*, Elsevier.

Table 10-1
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Cellular compartments





Cellular compartments

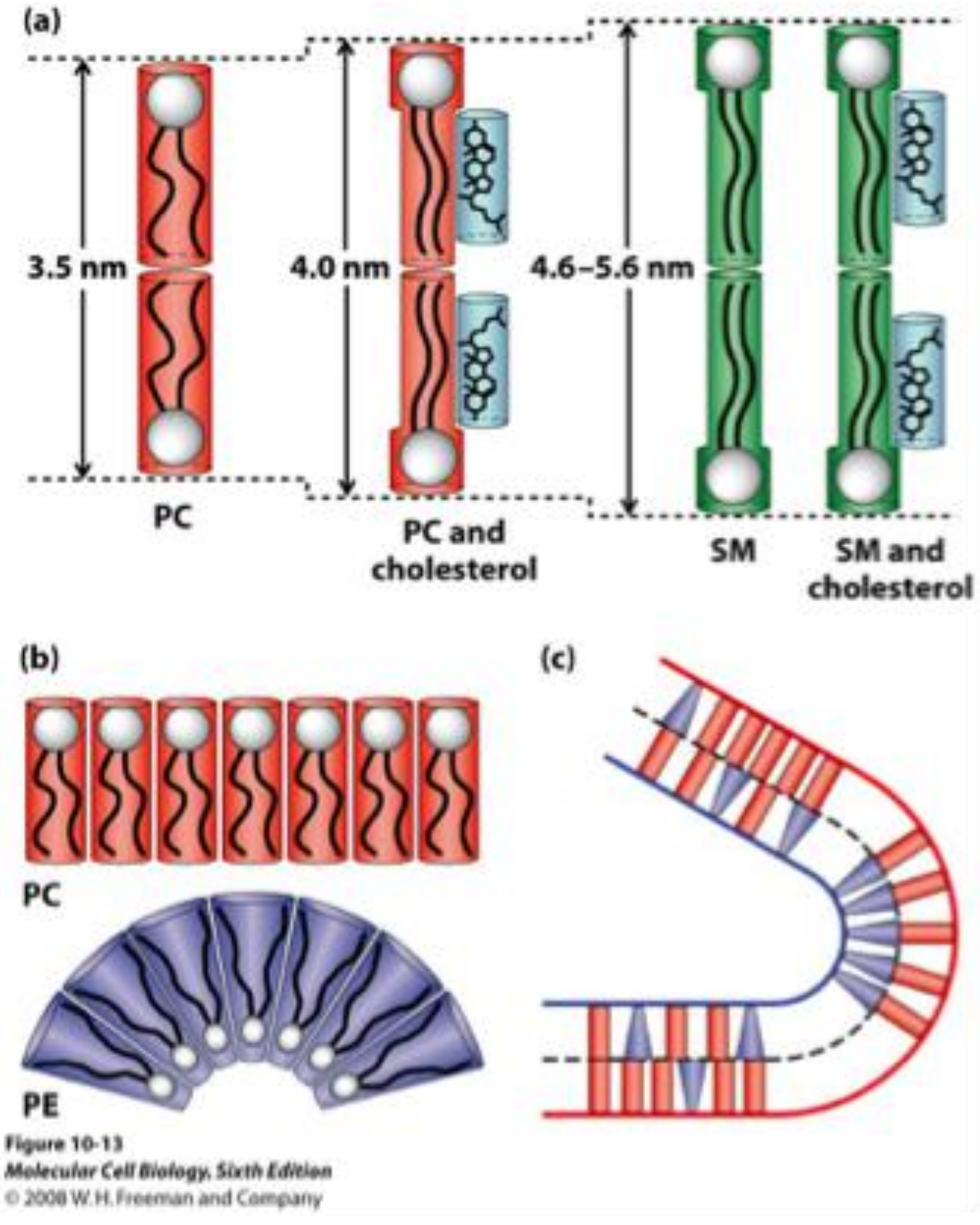
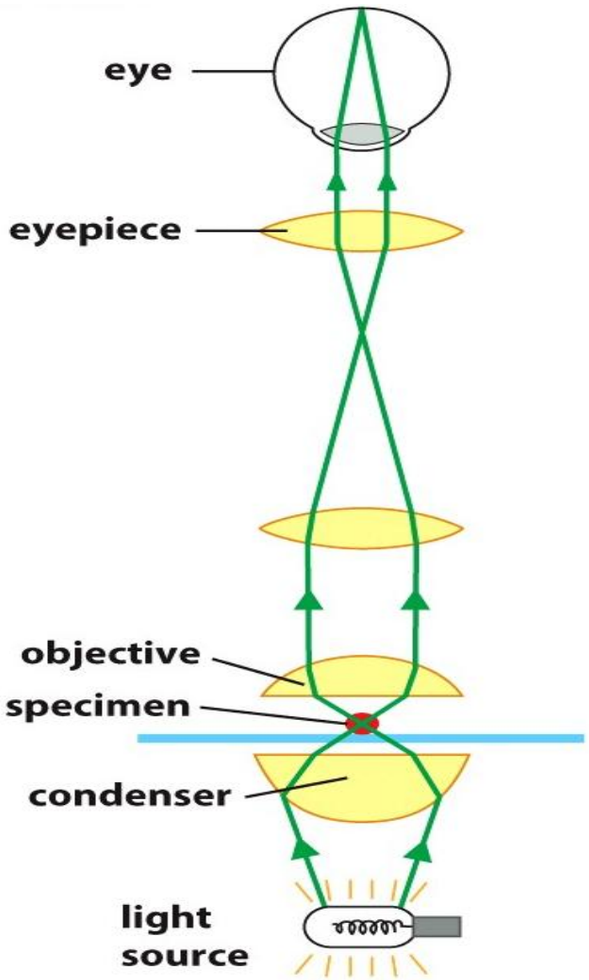


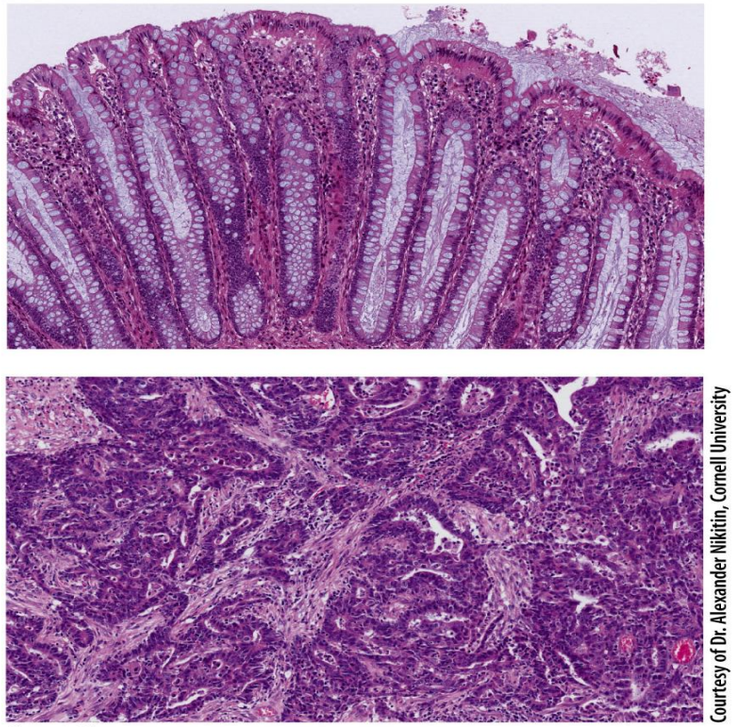
Figure 10-13
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Light Microscopy (Bright field)-Looking at Living Cells



- Haematoxylin and Eosin (H&E) stain.
- Haematoxylin binds to acidic molecules such as DNA and acidic amino acids such as aspartate and glutamate.
- Eosin binds to basic amino acids(lysine and arginine).

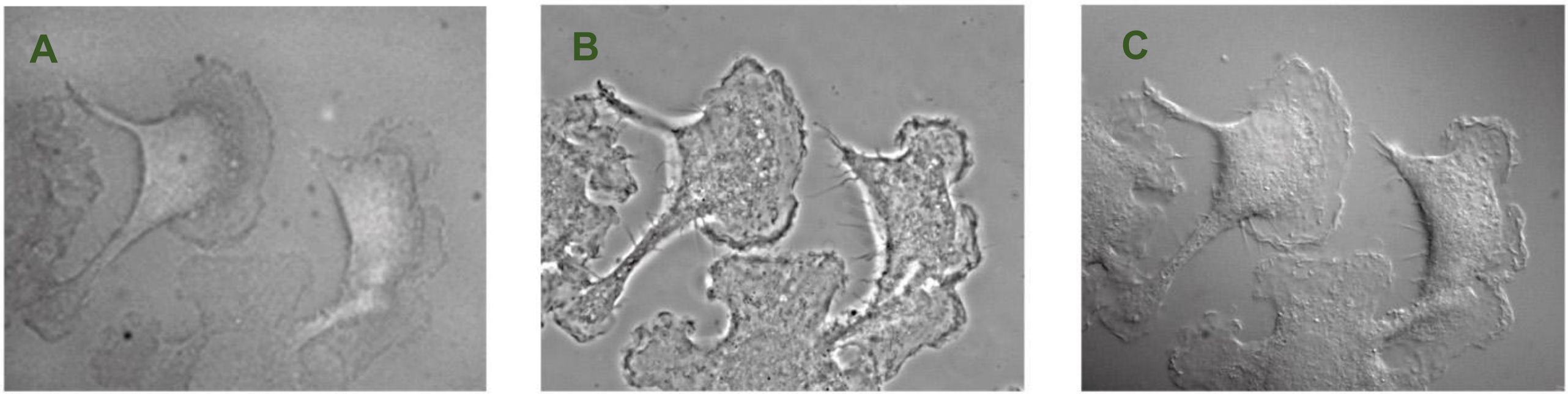
Normal human colon



Courtesy of Dr. Alexander Nikitin, Cornell University

Figure 4-11b
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Cancerous human colon



Courtesy of N. Watson and James Evans

Fluorescent dyes in light microscopy

Cultured cells are fixed (so as to preserve the structure and form) and permeabilised (so as to allow the dye to access the target). They are then incubated with fluorescently labelled proteins that bind to defined targets in the cell.

A

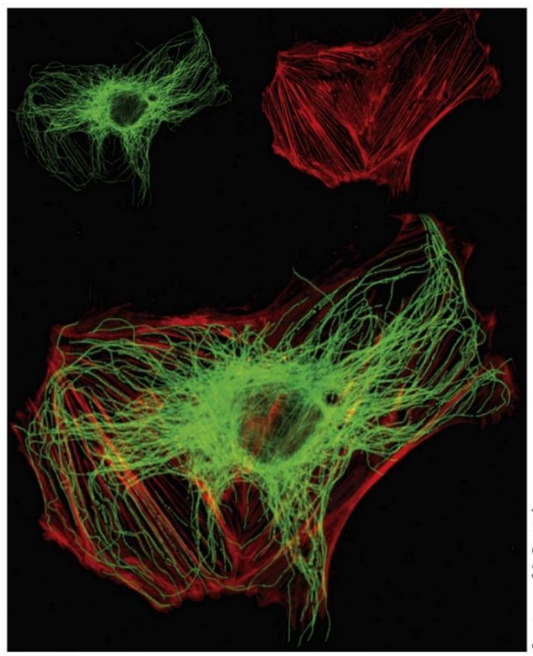


Figure 4-15b
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Fluorescently labelled cells showing microfilaments in red and tubulin green

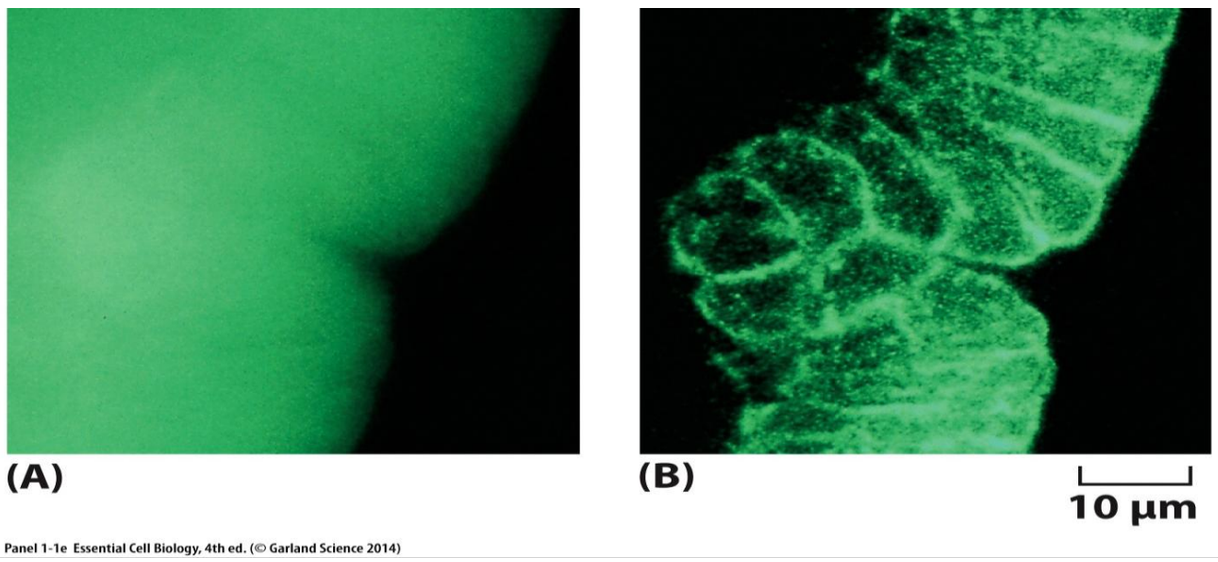
- Phalloidin is labelled with Rhodamine, a red fluorescent dye and binds to actin filaments.
- The antibody that binds to tubulin is labelled with a green fluorescent dye (fluorescein) in this example.
- The bottom panel shows both images superimposed

Confocal microscopy



Figure 4-19
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B



Panel 1-1e Essential Cell Biology, 4th ed. (© Garland Science 2014)

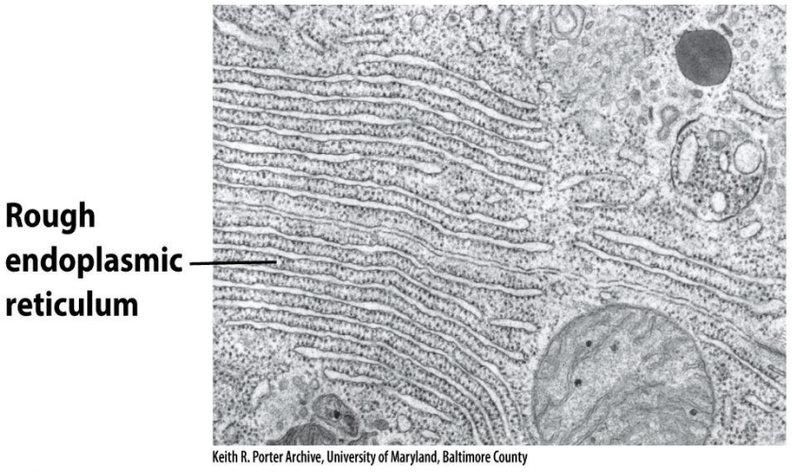
Intact insect embryo viewed with:
(a) Conventional microscopy
(b) Confocal microscopy

Electron Microscopy imaging

- Electron irradiation causes breakage of chemical bonds and creates free radicals that can cause further damage.
- Coating the specimens with heavy metals protects against this, but reduces the resolution,
- Cryo EM reduces the extent of this damage by about 6-fold and allows resolution of 2-3 Angstroms. (0.2-0.3nm).

Transmission EM

A



Keith R. Porter Archive, University of Maryland, Baltimore County

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Scanning EM

B

Steve Gschmeissner/Science Source

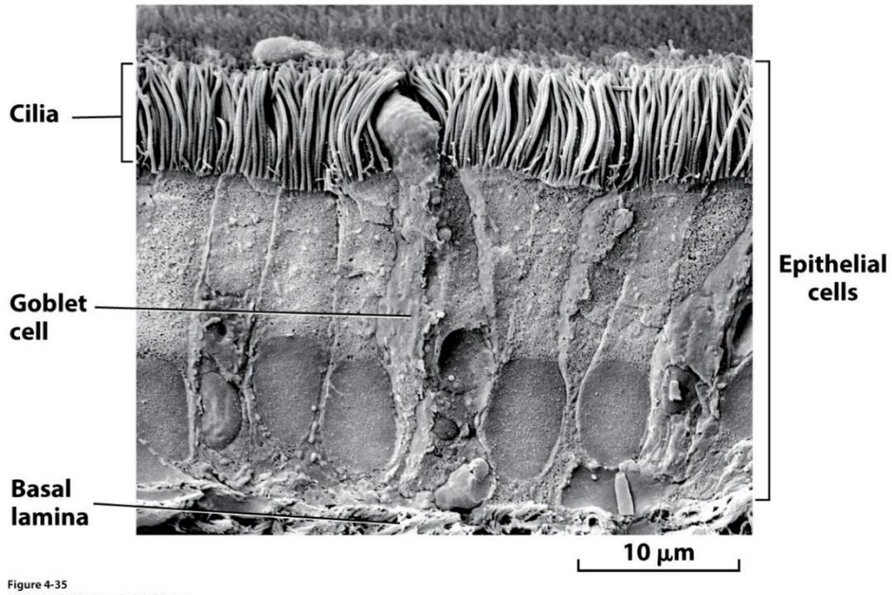
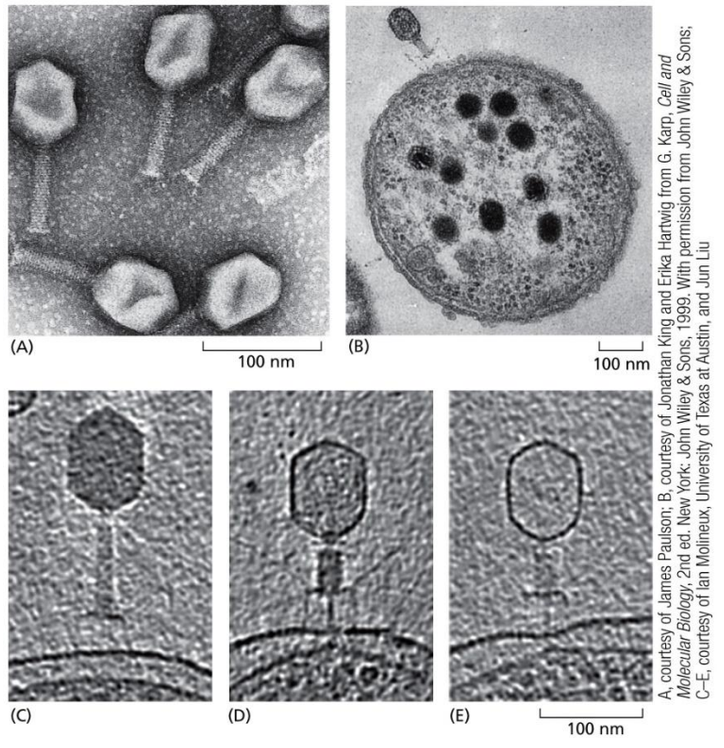


Figure 4-35
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Trachea with goblet cell in the middle and ciliated epithelial cells on either side.

Cryo(genic) EM

C



A, courtesy of James Paulson; B, courtesy of Jonathan King and Erika Hartwig from G. Karp, *Cell and Molecular Biology*, 2nd ed. New York. John Wiley & Sons, 1999. With permission from John Wiley & Sons; C–E, courtesy of Ian Molineux, University of Texas at Austin, and Jun Liu

Imaging using electron microscopy at very low temperatures of liquid nitrogen or liquid helium.

A: T4 bacteriophage
B:E.coli cross section with T4 attached on the surface and assemblies of T4 heads shown as black objects
C: T4 attachment to bacterium begins
D: Attached T4 during DNA injection
E: Injection complete-the T4 head is empty

Fig. 1-37 Molecular biology of the cell 7e Alberts et al

Take home message

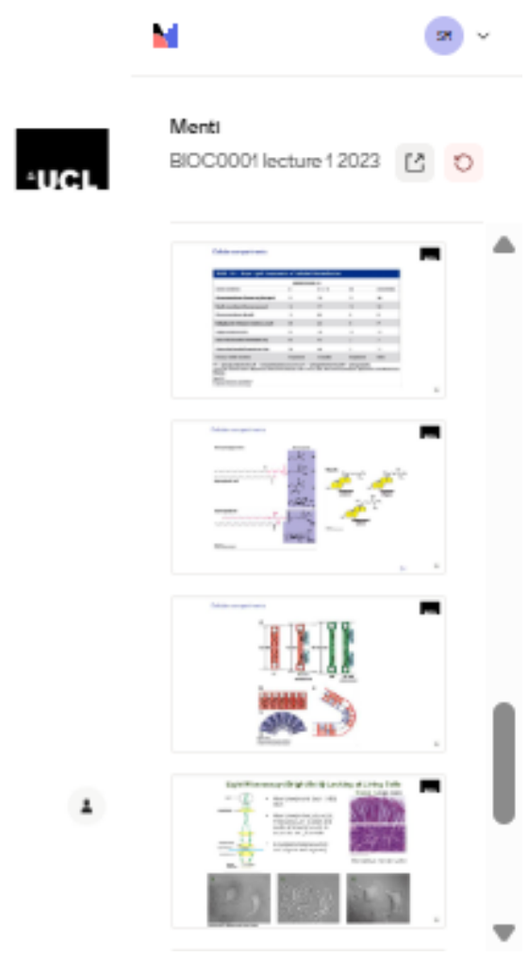
- Cells can be autonomous or part of a larger unit
- All life originates from a single ancestor
- Conservation illustrates natures way of adapting or solving a common problem
- Prokaryotes and eukaryotes differ!
- Membranes are integral to the cell or organelle but are fluid in composition
- Microscopy of differing resolutions allows us to see different perspectives in the cell.

BIOC0001

Lecture 1

Introduction to Cell Biology

Dr. Suzanne Ruddy




Have
a
Break
Waage
24.09 23.10